

Supplementary Material for *Credible on Paper? Evaluating the Effect of Green Banking Policy on Firms' Green Investment in Emerging Economies*

This file reports material referred to in the main text. Tables S1 and S2 give variable definitions and the composition of the primary sample. Table S3 reports the causal forest's conditional effects by firm-size category. Table S4 describes the small-sample check on waste-minimisation adoption, and Tables S5 and S6 report additional robustness checks and the multiple-testing correction. Table S7 gives the SBFN and regulatory-quality coding for every primary and supplementary economy, Section S7 the Bayesian convergence diagnostics, and Table S8 the global sample's SBFN membership by region. Main-text tables are referred to by description rather than by number.

S1 Variable definitions

Table S1 lists every variable used in the firm-level analysis, with its WBES question code and source.

Table S1: Variable definitions and sources.

Variable	Definition	WBES code	Source
green_adoption_binary	=1 if firm adopted ≥ 1 of 7 Green Economy practices in last 3 years	BMGc23b, c23d, c23e, c23f, c23j, c25, e5	WBES Green Economy Module (2018–2020)
fin_has_credit_line	=1 if firm has a line of credit or loan	K8	WBES Access to Finance Module
sbfn_member	=1 if SBFN policy adopted as of survey year	n/a	SBFN Global Progress Reports; data.sbfnetwork.org
wgi_regulatory_quality	WGI Regulatory Quality estimate, survey year	n/a	World Bank DataBank API (GOV_WGI_RQ_EST)
size_cat	Firm size stratum (Small/Medium/Large)	A6A	WBES Sampling Frame
sector_broad	Broad sector (Manufacturing/Services/Construction)	A4AX	WBES Sampling Frame
log_sales	Log of total annual sales	D2	WBES General Information Module
foreign_owned_dummy	=1 if any foreign ownership share $>0\%$	B2B	WBES Ownership Module
exporter_dummy	=1 if any export sales share $>0\%$	D3B, D3C	WBES Sales Module

S2 Sample composition

Table S2 reports, for each of the 41 primary-sample economies, the number of firms, survey year, SBFN status, regulatory quality, and unconditional rates of green-practice adoption and bank credit.

Table S2: Sample composition, by economy.

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Albania 2019	377	2019	0	0.09	0.609	0.407
Armenia 2020	546	2020	0	0.21	0.684	0.482
Azerbaijan 2019	225	2019	0	-0.10	0.418	0.180
Belarus 2018	600	2018	0	-0.79	0.662	0.425
Bosnia and Herzegovina 2019	362	2019	0	-0.38	0.581	0.565
Bulgaria 2019	772	2019	0	0.46	0.604	0.467
Croatia 2019	404	2019	0	0.38	0.708	0.448
Cyprus 2019	240	2019	0	1.04	0.746	0.711
Czechia 2019	502	2019	0	1.12	0.725	0.526
Egypt 2020	3,075	2020	1	-0.64	0.303	0.065
Estonia 2019	360	2019	0	1.54	0.738	0.412
Georgia 2019	581	2019	1	0.88	0.441	0.501
Greece 2018	600	2018	0	0.29	0.825	0.499
Hungary 2019	805	2019	0	0.46	0.673	0.537
Italy 2019	760	2019	0	0.70	0.506	0.230
Jordan 2019	601	2019	1	0.16	0.471	0.248
Kazakhstan 2019	1,446	2019	0	0.11	0.519	0.257
Kosovo 2019	271	2019	0	-0.38	0.698	0.425
Kyrgyz Republic 2019	360	2019	1	-0.42	0.606	0.299
Latvia 2019	359	2019	0	1.00	0.824	0.386
Lebanon 2019	532	2019	0	-0.56	0.452	0.365
Lithuania 2019	358	2019	0	0.98	0.678	0.342
Malta 2019	242	2019	0	0.83	0.901	0.479
Moldova 2019	360	2019	0	-0.11	0.619	0.328
Mongolia 2019	360	2019	1	-0.07	0.625	0.492
Montenegro 2019	150	2019	0	0.29	0.387	0.576
Morocco 2019	1,096	2019	1	0.03	0.478	0.251
North Macedonia 2019	360	2019	0	0.19	0.572	0.508
Poland 2019	1,369	2019	0	0.83	0.569	0.403
Portugal 2019	1,062	2019	0	1.00	0.812	0.600
Romania 2019	814	2019	0	0.36	0.438	0.474
Russia 2019	1,323	2019	0	-0.35	0.503	0.310
Serbia 2019	361	2019	0	0.06	0.654	0.585
Slovak Republic 2019	429	2019	0	0.73	0.782	0.386
Slovenia 2019	409	2019	0	0.88	0.793	0.674
Tajikistan 2019	352	2019	0	-0.80	0.523	0.194
Tunisia 2020	615	2020	1	-0.11	0.391	0.413
Türkiye 2019	1,663	2019	1	-0.02	0.346	0.345
Ukraine 2019	1,337	2019	0	-0.29	0.715	0.244
Uzbekistan 2019	1,239	2019	0	-0.92	0.751	0.240

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
West Bank and Gaza 2019	365	2019	0	-0.15	0.428	0.178

Note: Reg. qual. is the WGI Regulatory Quality estimate for the survey year. Green rate is the unconditional share of firms reporting adoption of at least one Green Economy practice. Credit rate is the share of firms reporting a bank credit line or loan.

S3 Causal forest effects by firm size

Table S3 reports the causal forest’s conditional average treatment effects of bank credit by the Enterprise Surveys’ firm-size strata. The estimates show no monotonic gradient, and all intervals overlap and include zero.

Table S3: Causal forest CATEs by firm-size category.

Size category	CATE	95% CI low	95% CI high
Small	0.0797	-0.0409	0.2004
Medium	0.0747	-0.0344	0.1838
Large	0.0569	-0.0428	0.1565

S4 Small-sample supplementary check: waste-minimisation adoption

Before the global-sample replication that is the paper’s second major test, we report a smaller supplementary check using waste-minimisation adoption—an outcome not captured in the standardised cross-country database underlying the global sample, and so a genuinely independent (if small-sample) data point—available in four economies whose Green Economy modules asked this item under directly comparable wording: India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021. Three of these four are themselves SBFN members at various progression stages, so this sample cannot re-test the SBFN-adoption contrast; what it can test is whether the credit-to-green channel documented in the main text’s baseline, hierarchical, and causal-forest results is a peculiarity of the ECA-MENA sample or generalises elsewhere. It generalises: pooled across the four economies, having an overdraft facility is associated with significantly higher odds of waste-minimisation adoption ($b = 0.422$, $p < 0.001$, $N = 9,038$, 4 countries). The direction of the core finance-green relationship replicates on a different continent, a different half-decade, and a narrower outcome definition than the one anchoring the main text’s primary-sample results—a preview, on four economies, of the much larger replication in the global sample.

Table S4: Small-sample supplementary check: descriptive rates.

Economy	N	Waste-min. rate	Overdraft rate	SBFN member
India 2022	9,376	0.515	0.621	1
Indonesia 2023	2,955	0.336	0.144	1
Peru 2023	987	0.806	0.549	1
Timor-Leste 2021	238	0.061	0.089	0

S5 Additional robustness checks

Table S5 subjects the primary-sample finding to four further checks, all retaining sector fixed effects, firm controls, and country-clustered standard errors (no country fixed effects, for the reason given in the main text). First, replacing the binary outcome with the continuous seven-item green-adoption index and estimating by OLS reproduces the same pattern to the decimal: credit access positive and significant ($b = 0.053$, $p < 0.001$), SBFN main effect negative and significant ($b = -0.092$, $p < 0.001$), interaction indistinguishable from zero ($b = 0.006$, $p = 0.739$).

Second, replacing bank credit access with two alternative finance measures produces one genuine point of nuance worth reporting rather than smoothing over. Using overdraft-facility access instead of a credit line/loan, the main overdraft effect remains positive and significant ($b = 0.274$, $p = 0.011$), the SBFN main effect remains negative and significant ($b = -0.905$, $p < 0.001$)—but the overdraft $\times$ SBFN interaction is now negative and significant ($b = -0.346$, $p = 0.031$), the one interaction term in the entire paper that clears a conventional significance threshold, and it runs opposite to the direction H2 predicts. Using the reverse-coded perceived finance-obstacle measure in place of realised access, by contrast, both the main effect and its SBFN interaction are precisely zero ($b = -0.010$, $p = 0.827$; interaction $b = -0.012$, $p = 0.849$)—consistent with the descriptive correlation noted in the main text. We read the overdraft result as suggestive rather than conclusive given it is one significant coefficient among more than a dozen interaction tests across the paper (see Table S6 below), but it is at least directionally consistent with a substantive story worth flagging for future work: if SBFN-guided lending is channelled preferentially through the term loans and credit lines that finance long-horizon capital expenditure, rather than through short-term overdraft facilities, we might expect exactly this pattern—a null-to-positive interaction on credit lines and a negative one on overdrafts—though our data cannot directly test whether SBFN-guided credit is disproportionately term-structured.

Third, splitting the sample into manufacturing ($N = 13,340$) and services ($N = 9,538$) subsamples shows the credit-access main effect, the SBFN main effect, and the null credit $\times$ SBFN interaction all replicate within each sector separately (manufacturing: credit $b = 0.480$, $p < 0.001$, interaction $b = -0.040$, $p = 0.826$; services: credit $b = 0.448$, $p < 0.001$, interaction $b = 0.195$, $p = 0.426$)—the paper’s central null is not an artefact of pooling two structurally different production technologies.

Because the overdraft interaction (the one coefficient in the paper that clears a conventional significance threshold) is one of many credit $\times$ institutional-moderator tests reported across the frequentist specifications in this paper, we additionally apply a Benjamini–Hochberg false-discovery-rate correction across the full family of 13 such coefficients reported

Table S5: Additional robustness checks.

Specification	Finance coef. (p)	SBFN coef. (p)	Interaction coef. (p)
(a) Continuous index, OLS	0.053 (0.000)	-0.092 (0.000)	0.006 (0.739)
(b1) Overdraft	0.274 (0.011)	-0.905 (0.000)	-0.346 (0.031)
(b2) Finance obstacle (reverse-coded)	-0.010 (0.827)	-1.011 (0.000)	-0.012 (0.849)
(c1) Manufacturing subsample	0.480 (0.000)	-1.047 (0.000)	-0.040 (0.826)
(c2) Services subsample	0.448 (0.000)	-0.914 (0.001)	0.195 (0.426)

under a classical (non-Bayesian, non-forest) estimator in the main text’s baseline, robustness, and global-sample regression tables (Table S6 below; full calculation in the replication package). The overdraft interaction’s raw $p = 0.031$ becomes $p = 0.403$ after correction, and no coefficient in the family survives at a false discovery rate of 0.10 or even 0.40. We read this as reinforcing, with a formal multiple-testing accounting rather than only a qualitative caveat, the suggestive-not-conclusive framing the main text’s Discussion already gives this result: an isolated nominally-significant interaction among more than a dozen tested is exactly what a pure false-discovery process would be expected to produce roughly one time in twenty, and this correction does not let us reject that account.

Table S6: Multiple-testing correction: Benjamini–Hochberg false discovery rate applied to every credit $\times$ institutional-moderator interaction coefficient reported under a frequentist estimator (main text’s baseline, robustness, and global-sample regression tables above). Entries marked $\dagger$ are approximate p -values recovered from the reported coefficient and standard error via a normal approximation ($z = \text{coef}/\text{se}$), since the main text reports only the coefficient and standard error for these terms.

Source	Term	Raw p	BH-adjusted p
Table S5 (b1) Overdraft	Overdraft $\times$ SBFN	0.031	0.403
Main text, global sample, M2 (country FE)	Credit $\times$ SBFN	0.205	0.799
Main text, global sample, M1 (no country FE)	Credit $\times$ Reg. quality †	0.301	0.799
Main text, baseline M3 †	Credit $\times$ Reg. quality	0.310	0.799
Main text, global sample, M2 (country FE)	Credit $\times$ Reg. quality	0.346	0.799
Table S5 (c2) Services	Credit $\times$ SBFN	0.426	0.799
Main text, global sample, M1 (no country FE)	Credit $\times$ SBFN	0.430	0.799
Main text, baseline M3 †	Credit $\times$ SBFN $\times$ Reg. quality	0.702	0.892
Table S5 (a) Continuous index, OLS	Credit $\times$ SBFN	0.739	0.892
Main text, baseline M2	Credit $\times$ SBFN	0.773	0.892
Table S5 (c1) Manufacturing	Credit $\times$ SBFN	0.826	0.892
Table S5 (b2) Finance obstacle (reverse-coded)	Obstacle $\times$ SBFN	0.849	0.892
Main text, baseline M3 †	Credit $\times$ SBFN	0.892	0.892

S6 SBFN policy status and regulatory quality, full sample

Table S7 reports, for all 47 economy-years in the paper’s primary and small-sample supplementary check, the SBFN membership status coded for the analysis, the join year where

applicable, the SBFN Multi-Tiered Progress Framework stage in force nearest the survey year, and the exact year of the Worldwide Governance Indicators observation used. Full per-country source citations (the specific SBFN Global Progress Report, country addendum, or accession announcement consulted for every determination) are available in the replication package. The analogous 162-economy table for the global sample is summarised by region in Table S8 below and provided in full in the replication package.

Table S7: SBFN policy status and regulatory quality, full 47-economy sample.

Economy	SBFN member	Join year	Stage at survey	WGI year
Albania 2019	0	—	None	2019
Armenia 2020	0	—	None	2020
Azerbaijan 2019	0	—	None	2019
Bangladesh 2022	1	2012	Advancing (ESG Integration)	2022
Belarus 2018	0	—	None	2018
Bosnia and Herzegovina 2019	0	—	None	2019
Bulgaria 2019	0	—	None	2019
Croatia 2019	0	—	None	2019
Cyprus 2019	0	—	None	2019
Czechia 2019	0	—	None	2019
Egypt 2020	1	2016	Formulating (Preparation)	2020
Estonia 2019	0	—	None	2019
Georgia 2019	1	2017	Developing (Implementation)	2019
Greece 2018	0	—	None	2018
Hungary 2019	0	—	None	2019
India 2022	1	2016	Developing (Commitment)	2022
Indonesia 2023	1	2012	Consolidating (Maturing)	2023
Italy 2019	0	—	None	2019
Jordan 2019	1	2016	Commitment (Preparation)	2019
Kazakhstan 2019	0	—	None	2019
Kosovo 2019	0	—	None	2019
Kyrgyz Republic 2019	1	2018	Commitment (Preparation)	2019
Latvia 2019	0	—	None	2019
Lebanon 2019	0	—	None	2019
Lithuania 2019	0	—	None	2019
Malta 2019	0	—	None	2019
Moldova 2019	0	—	None	2019
Mongolia 2019	1	2012	Advancing (Implementation)	2019
Montenegro 2019	0	—	None	2019

Economy	SBFN member	Join year	Stage at survey	WGI year
Morocco 2019	1	2014	Advancing (Implementa- tion)	2019
North Macedo- nia 2019	0	—	None	2019
Peru 2023	1	2013	Advancing (ESG Integra- tion)	2023
Philippines 2023	1	2013	Advancing (ESG Integra- tion)	2023
Poland 2019	0	—	None	2019
Portugal 2019	0	—	None	2019
Romania 2019	0	—	None	2019
Russia 2019	0	—	None	2019
Serbia 2019	0	—	None	2019
Slovak Repub- lic 2019	0	—	None	2019
Slovenia 2019	0	—	None	2019
Tajikistan 2019	0	—	None	2019
Timor-Leste 2021	0	—	None	2021
Tunisia 2020	1	2019	Unverified	2020
Türkiye 2019	1	2015	Developing (Implementa- tion)	2019
Ukraine 2019	0	—	None	2019
Uzbekistan 2019	0	—	None	2019
West Bank and Gaza 2019	0	—	None	2019

S7 Bayesian model convergence diagnostics

Convergence diagnostics for every focal parameter of the hierarchical model reported in the main text are included directly in that table ($\hat{R}$ column); the full posterior summary also includes ess_{bulk} ranging from 797 (SBFN member) to 2,270 (credit $\times$ regulatory quality), comfortably above the conventional minimum threshold of 400 effective samples per parameter for stable posterior summaries at four chains. No divergent transitions were flagged during sampling (NUTS via the nutpie sampler, four chains of 1,000 post-warmup draws each after 1,000 tuning iterations).

S8 Global-sample SBFN status, regional summary

Of the 162 global-sample economy-years, 61 are coded as SBFN members as of their survey year and 101 as non-members. Table S8 reports the regional distribution of the 61 members, drawn from the SBFN data portal’s regional classification; five of the 61 (Peru, Philippines, Timor-Leste, Bangladesh, and Indonesia) carry the deeper per-country sourcing already reported in Table S7 above, having been determined during the primary sample’s original

research pass rather than from the portal roster alone.

Table S8: Global-sample SBFN members, by SBFN region.

Region	Member economy-years
Latin America and Caribbean	17
Africa	12
Europe and Central Asia	11
East Asia and Pacific	11
South Asia	6
Middle East and North Africa	4
Total	61